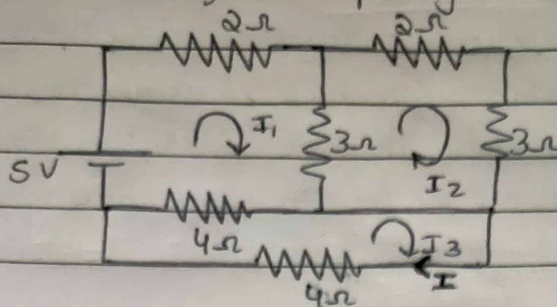


- 1) Find Current I & verify Reciprocity.



- Apply KVL to loop 1 [Using Matrix Method].

$$R_{11}I_1 + R_{12}I_2 + R_{13}I_3 = V$$

$$9I_1 - 3I_2 - 4I_3 = 5 \rightarrow (1)$$

- Apply KVL to loop 2.

$$R_{21}I_1 + R_{22}I_2 + R_{23}I_3 = V$$

$$-3I_1 + 8I_2 = 0 \rightarrow (2)$$

- Apply KVL to loop 3 $\Rightarrow R_{31}I_1 + R_{32}I_2 + R_{33}I_3 = V$

$$-4I_1 + 8I_2 = 0 \rightarrow (3)$$

Solving Eq. (1), (2) & (3)

$$I_1 = 0.85A, I_2 = 0.31A, I_3 = 0.42A$$

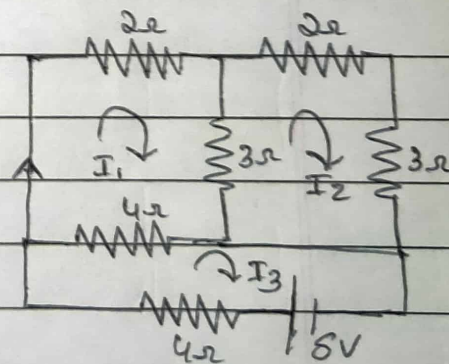
- Interchange the Excitation & Response.

- Apply KVL to loop 1

$$9I_1 - 3I_2 - 4I_3 = 0 \rightarrow (1)$$

- Apply KVL to loop 2

$$-3I_1 + 8I_2 = 0 \rightarrow (2)$$



- Apply KVL to loop 3

$$-4I_1 + 8I_3 = 5 \rightarrow (3)$$

Solve (1), (2) & (3)

$$I = 0.425, I_2 = 0.159A, I_3 = 0.837A$$

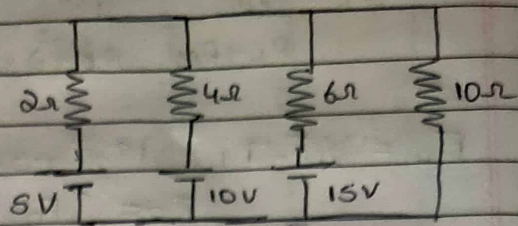
$$I = I_1 = I_3 = 0.425$$

2) Find Current through 10Ω using Millman's theorem

$$\rightarrow V_m = \frac{V_1 G_1 + V_2 G_2 + V_3 G_3}{G_1 + G_2 + G_3}$$

$$= \frac{5/2 + 10/4 + 6/15}{1/5 + 1/4 + 1/15}$$

$$= \frac{15/2}{0.916} = 8.18 \text{ V} = V_m$$

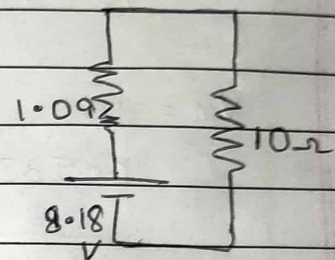


$$\bullet R_m = \frac{1}{G_m} = \frac{1}{G_1 + G_2 + G_3} = \frac{1}{1/5 + 1/4 + 1/6} = 1.09 \Omega = R_m$$

• Eqn ckt

$$\text{load current } I_L = \frac{V_m}{R_m + R_L} = \frac{8.18}{1.09 + 10}$$

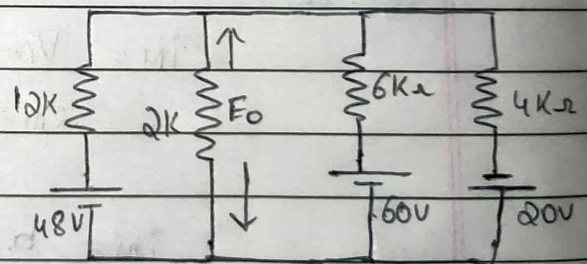
$$I_L = 0.737 \text{ A}$$



*4). Find the output Voltage E_o using Millman's Theorem.

$$\rightarrow V_m = \frac{V_1 G_1 + V_2 G_2 - V_3 G_3}{G_1 + G_2 + G_3}$$

[The ~~1st~~ source is in '-' direction
So the ~~sgn~~ is $[-V_3 G_3]$]

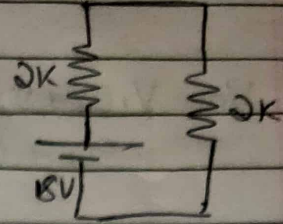


$$V_m = \frac{48/12 \times 10^3 + 60/6 \times 10^3 - 20/4 \times 10^3}{1/12 \times 10^3 + 1/6 \times 10^3 + 1/4 \times 10^3} = \frac{9 \times 10^{-3}}{5 \times 10^{-4}} = 18 \text{ V} = V_m$$

$$R_m = \frac{1}{G_m} = \frac{1}{G_1 + G_2 + G_3} = \frac{1}{1/12 \times 10^3 + 1/6 \times 10^3 + 1/4 \times 10^3} = 2 \text{ k}\Omega = R_m$$

Ex ① Ckt

$$I_L = \frac{V_m}{R_L + R_M} = \frac{18}{2 \times 10^3 + 2 \times 10^3} = 4.5 \text{ mA} = I_L$$



~~$$E_o = V_m \cdot I_L = 18 \times 4.5 \times 10^{-3}$$~~

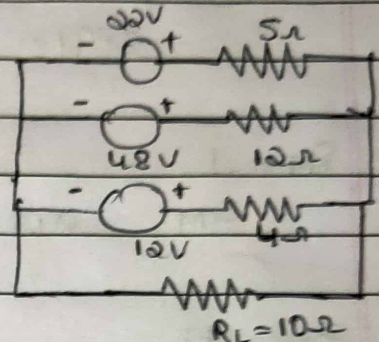
$$E_o = I_L \times R_L = 4.5 \times 10^{-3} \times 2 \times 10^3$$

$$E_o = 9 \text{ V}$$

5) Using Millman's theorem find Current through load Resistor $R_L = 100\Omega$

$$\rightarrow V_m = \frac{V_1 G_1 + V_2 G_2 + V_3 G_3}{G_1 + G_2 + G_3}$$

$$V_m = \frac{20/5 + 48/12 + 12/4}{1/5 + 1/12 + 1/4}$$



$$V_m = 11.4 = 21.38 \text{ V} = V_m$$

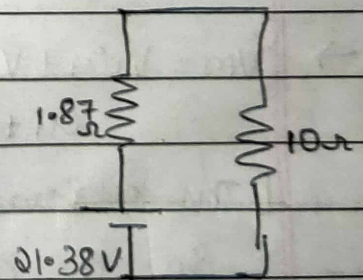
$$0.533$$

$$R_m = \frac{1}{G_m} = \frac{1}{G_1 + G_2 + G_3} = \frac{1}{1/5 + 1/12 + 1/4} = \frac{1}{0.533} = 1.87\Omega = R_m$$

Ex ② Ckt

$$i_m = \frac{V_m}{R_m + R_L}$$

$$i_m = \frac{21.38}{1.87 + 10}$$

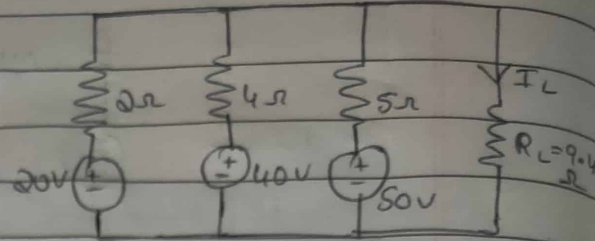


$$i_m = 1.8 \text{ A}$$

7) ~~Find the current through R_L using Millman's theorem~~ Using Millman's theorem find current through R_L .

$$\rightarrow V_M = \frac{V_1 G_1 + V_2 G_2 + V_3 G_3}{G_1 + G_2 + G_3}$$

$$V_M = \frac{20/2 + 40/4 + 50/5}{1/2 + 1/4 + 1/5}$$



$$\bullet V_M = \frac{30}{0.95} = 31.57V = V_M //$$

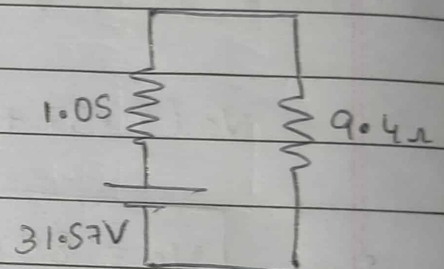
$$\bullet R_M = \frac{1}{\frac{1}{2} + \frac{1}{4} + \frac{1}{5}}$$

$$R_M = \frac{1}{0.95} = 1.05\Omega = R_M //$$

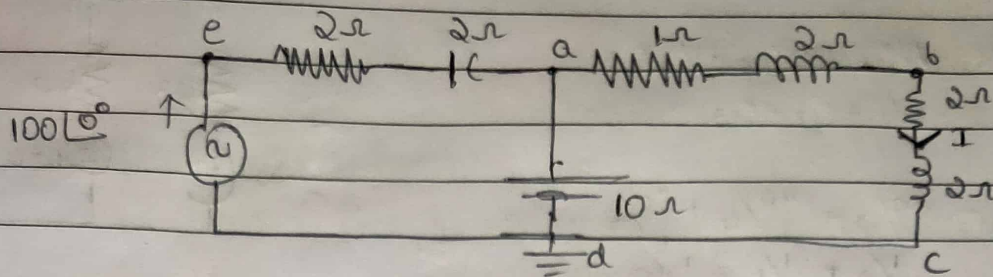
• ~~Eq~~ I_L

$$I_L = \frac{V_M}{R_M + R_L} = \frac{31.57}{1.05 + 9.4}$$

$$I_L = 3.02A //$$



3) Verify Reciprocity theorem by finding current through branch b c



→
Case i

Using Nodal Analysis

Apply KCL at Node A

Capacitor = $-j2$

inductor = $j2$

$$\rightarrow \frac{V_a - 100\angle 0^\circ}{2 - j2} + \frac{V_a}{-j10} + \frac{V_a - V_c}{3 + j4} = 0 \quad [V_c = 0 \text{ ground potential}]$$

$$V_a \left(\frac{1}{2 - j2} + \frac{1}{-j10} + \frac{1}{3 + j4} \right) = \frac{100}{2 - j2}$$

$$V_a (0.37 + 0.19j) = 25 + 25j$$

$$\bullet \quad \sqrt{V_a = 80.92 + 26.01j} \Rightarrow [V_a = 85 \angle 17.82^\circ]$$

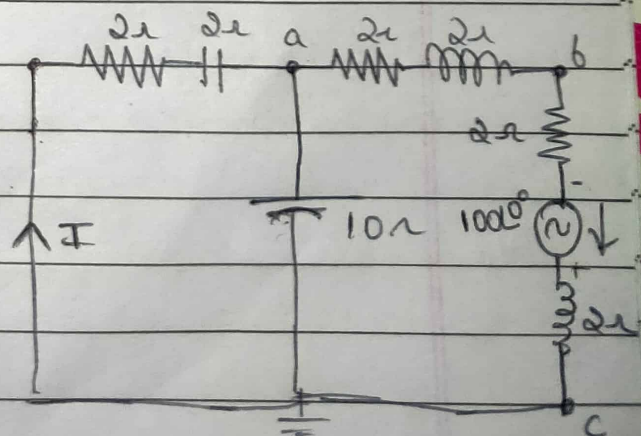
$$\bullet \quad I = \frac{V_a}{3 + j4} = \frac{85 \angle 17.82^\circ}{5 \angle 53.13^\circ} = 17 \angle -35.31^\circ$$

• Case ii Interchange position of V & I

Apply KCL at Node A

$$\frac{V_a}{2 - j2} + \frac{V_a}{-j10} + \frac{V_a + 100\angle 0^\circ}{3 + j4} = 0$$

$$V_a \left[\frac{1}{2 - j2} + \frac{1}{-j10} + \frac{1}{3 + j4} \right] = \frac{-100}{3 + j4}$$



$$V_a(0.37 + 0.19j) = -12 + j16$$

$$V_a = -8 + j47.3$$

$$|V_a| = 48.199.59$$

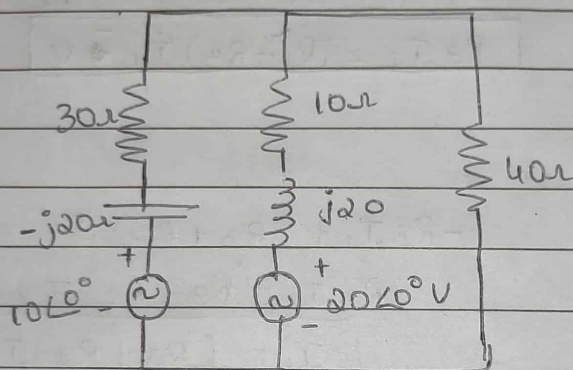
$$I = \frac{V_a}{2-j2} = \frac{48.199.59}{2.828 \angle 45^\circ} = 17 \angle 144.59^\circ$$

$$180 - 144.59$$

$$= +144.59 - 180 = -35.41$$

$$|I| = 17 \angle -35.41$$

8). Find the Current through 40Ω resistor for n/w using Millman's theorem



$$\text{Ans } V_M = \frac{10 \angle 0^\circ}{(30 - j20)} + \frac{20 \angle 0^\circ}{(10 + j20)} = 18.2 \angle -15.95^\circ \text{ V}$$

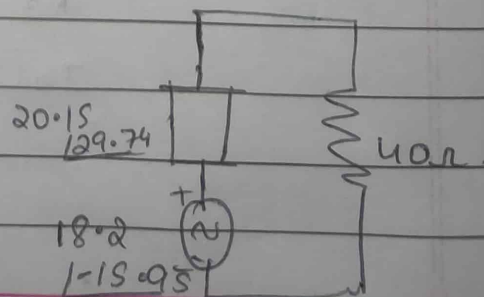
$$\frac{1}{(30 - j20)} + \frac{1}{10 + j20}$$

$$Z_M = \frac{1}{Y_M} = \frac{1}{(1/30 - j20) + (1/(10 + j20))} = 20.15 \angle 29.74^\circ \Omega$$

eg, n/w

$$I_L = \frac{18.2 \angle -15.95^\circ}{20.15 \angle 29.74^\circ + 40}$$

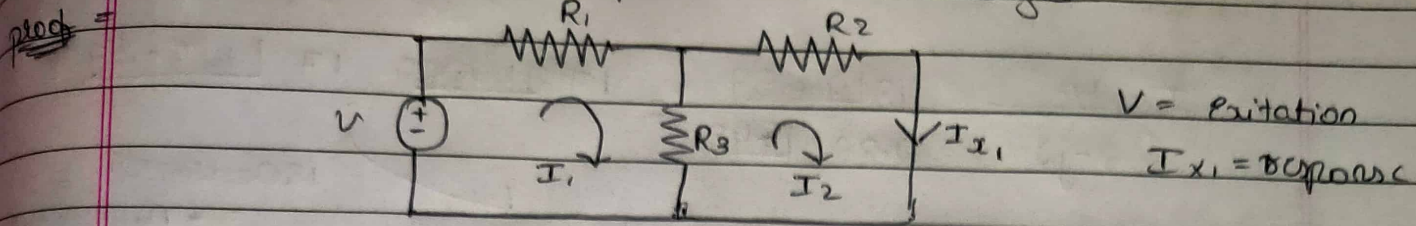
$$|I_L| = 0.31 \angle -25.81^\circ$$



$$0.281 - 0.126$$

6) State and Prove Reciprocity Theorem

→ In a linear, bilateral, active single source network, the ratio of Excitation to response remains same when the positions of Excitation and response are interchanged.



- Apply KVL to Mesh 1

$$(R_1 + R_3) I_1 - R_3 I_2 = V$$

$$(R_1 + R_3) I_1 - V = R_3 I_2$$

$$R_3 I_2 = (R_1 + R_3) I_1 - V \rightarrow \textcircled{1}$$

- Apply KVL to Mesh 2

$$-R_3 I_1 + (R_3 + R_2) I_2 = 0$$

$$R_3 I_1 = (R_3 + R_2) I_2$$

$$I_1 = \frac{(R_3 + R_2) I_2}{R_3} \rightarrow \textcircled{2}$$

Sub for I_1 in $\textcircled{1}$

$$R_3 I_2 = (R_1 + R_3) \left[\frac{(R_3 + R_2) I_2}{R_3} \right] - V$$

$$R_3 I_2 = \frac{(R_1 R_3 + R_1 R_2 + R_3^2 + R_2 R_3)}{R_3} I_2 - V$$

$$R_3^2 I_2 = (R_1 R_3 + R_1 R_2 + R_3^2 + R_2 R_3) I_2 - V R_3$$

$$I_2 (R_3^2 - R_1 R_3 - R_1 R_2 - R_3^2 - R_2 R_3) = -V R_3$$

$$-(R_1 R_3 + R_1 R_2 + R_2 R_3) I_2 = -V R_3$$

$$I_2 = \frac{V R_3}{(R_1 R_2 + R_2 R_3 + R_3 R_1)}$$

$$= I_{x1}$$

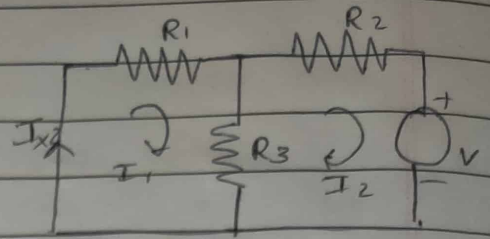
$$(R_1 R_2 + R_2 R_3 + R_3 R_1)$$

Inductance Excitation & Response:

Apply KVL for Mesh 1

$$(R_1 + R_3)I_1 - R_3 I_2 = 0$$

$$I_1 = \frac{R_3}{R_1 + R_3} I_2 \rightarrow (1)$$



Mesh 2 $-R_3 I_1 + (R_2 + R_3)I_2 = -V$

$$(R_2 + R_3)I_2 = -V + R_3 I_1$$

$$I_2 = \frac{-V + R_3 I_1}{R_2 + R_3} \rightarrow (2)$$

$$I_1 = \frac{R_3}{R_1 + R_3} \left[\frac{R_3 I_1 + V}{R_2 + R_3} \right]$$

$$I_1 = \frac{R_3^2 I_1}{(R_1 + R_3)(R_2 + R_3)} + \frac{VR_3}{(R_1 + R_3)(R_2 + R_3)}$$

$$I_1 \left[1 - \frac{R_3^2}{(R_1 + R_3)(R_2 + R_3)} \right] = \frac{VR_3}{(R_1 + R_3)(R_2 + R_3)}$$

$$I_1 \left[\frac{R_1 R_2 + R_1 R_3 + R_2 R_3 + \cancel{R_3^2} - \cancel{R_3^2}}{(R_1 + R_3)(R_2 + R_3)} \right] = \frac{VR_3}{(R_1 + R_3)(R_2 + R_3)}$$

$$\boxed{I_1 = \frac{VR_3}{R_1 R_2 + R_1 R_3 + R_2 R_3}} = I_{x2}$$

$$I_{x1} = I_{x2}$$

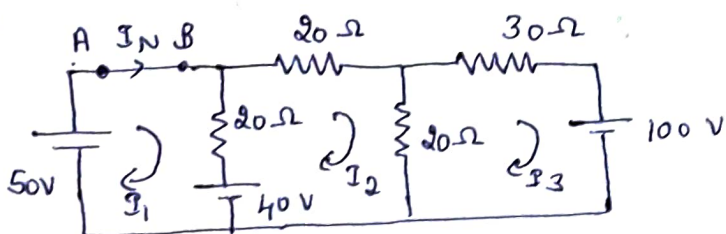
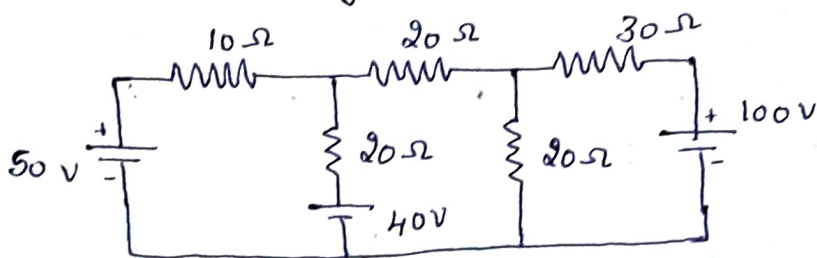
Network Analysis [BEC 24402]

Module - 2

Network Theorems

Unit - 2

Q1. Find the current through the 10Ω resistor using Norton's theorem.



Step 1:- A.S.C the load resistance and apply KVL to all 3 Mesh.

Mesh ①:- $50 - 20(I_1 - I_2) - 40 = 0$
 $50 - 20I_1 + 20I_2 - 40 = 0$
 $-20I_1 + 20I_2 = -10$ — (1)

Mesh ②:- $-20(I_2 - I_1) - 20I_2 - 20(I_2 - I_3) + 40 = 0$
 $-20I_2 + 20I_1 - 20I_2 - 20I_2 + 20I_3 = -40$
 $20I_1 - 60I_2 + 20I_3 = -40$ — (2)

Mesh ③:- $-30I_3 - 100 - 20(I_3 - I_2) = 0$
 $-30I_3 - 20I_3 + 20I_2 = 100$
 $20I_2 - 50I_3 = 100$ — (3)

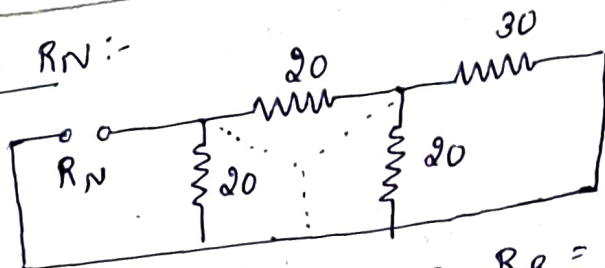
Solve eqn ①, ② & ③

$$I_1 = 0.8125 A, \quad I_2 = 0.3125 A, \quad I_3 = -1.875 A$$

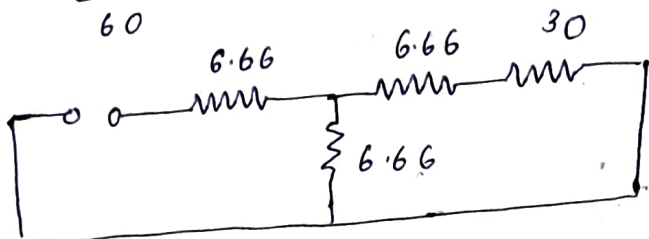
$$I_1 = I_N = 0.8125 A$$

Step 2:- S.C the voltage source and find the Norton's resistances.

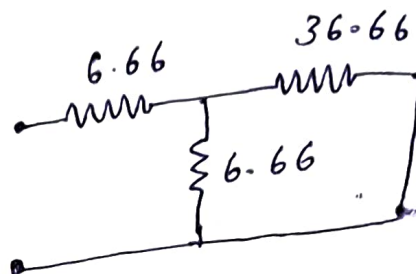
To find R_N :-



$$R_A = \frac{20 \times 20}{60} = 6.66 = R_B = R_C$$



$\Rightarrow$

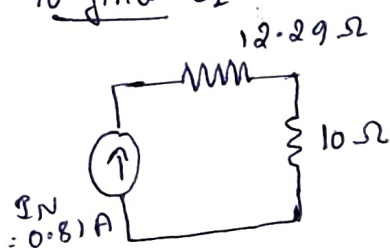


$$R_N = (6.66 \parallel 36.66) + 6.66$$

$$R_N = 12.29 \Omega$$

Step 3:- Find the equivalent circuit & find the load current $[I_L]$

To find I_L :-

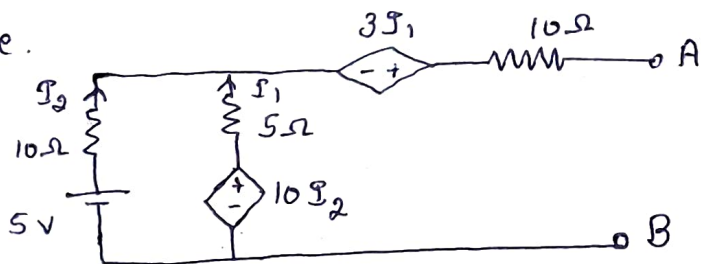


$$I_L = \frac{I_N \times R_N}{R_N + R_L}$$

$$= \frac{0.8125 \times 12.29}{12.29 + 10}$$

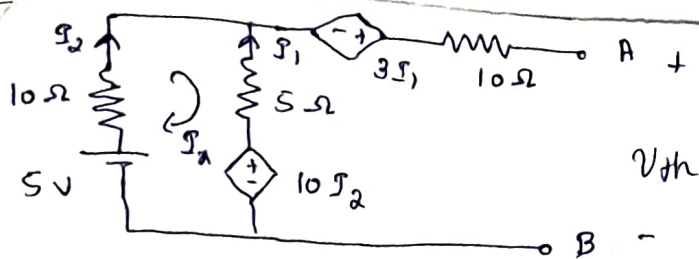
$$I_L = 0.44 \text{ A}$$

Q2. Find Norton's equivalent network across terminals A & B of the figure.



$\Rightarrow$ In the given above circuit it consists of Both dependent & Independent source so find V_{th} .

Step 1:- To find V_{th} .



To find current :-

$$I_x = I_2$$

$$I_1 = -I_x$$

$$5 - 10I_x - 5I_x - 10I_x = 0$$

$$-25I_x = -5$$

$$I_x = \frac{5}{25} = 0.2$$

$$I_x = 0.2 \text{ A}$$

$$I_1 = -0.2 \text{ A}, \quad I_2 = 0.2 \text{ A}$$

To find V_{th} :-

$$3I_1 + 0 - V_{th} + 10I_2 + 5I_x = 0$$

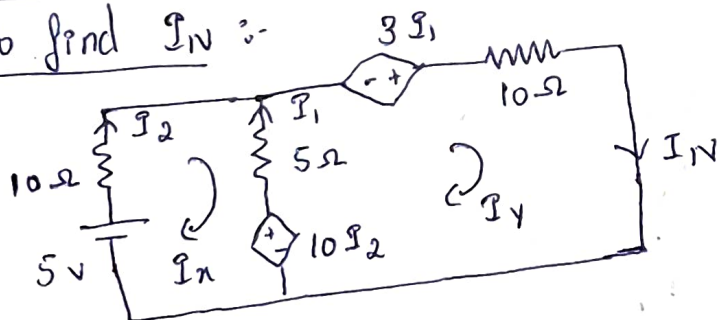
$$V_{th} = 3I_1 + 10I_2 + 5I_x$$

$$= -0.6 + 2 + 1$$

$$V_{th} = 2.4 \text{ V}$$

Step 2 :-

To find I_N :-



$$I_1 = I_y - I_x, \quad I_x = I_2$$

Apply KVL to Mesh

Mesh 1 :- $5 - 10I_x - 5(I_x - I_y) - 10I_2 = 0$

$$5 - 10I_x - 5I_x + 5I_y - 10I_x = 0$$

$$-25I_x + 5I_y = -5 \quad \text{--- (1)}$$

Mesh ② :- $3I_1 - 10I_4 + 10I_2 - 5(I_4 - I_N) = 0$
 $3I_1 - 10I_4 + 10I_2 - 5I_4 + 5I_N = 0$
 $3I_4 - 3I_N - 10I_4 + 10I_2 - 5I_4 + 5I_N = 0$
 $12I_N - 12I_4 = 0 \text{ --- (2)}$

Solving eqn (1) & (2) we get,

$$I_N = 0.25 \text{ A}$$

$$I_4 = 0.25 \text{ A}$$

$$I_N = I_4 = 0.25 \text{ A}$$

Step 3:-

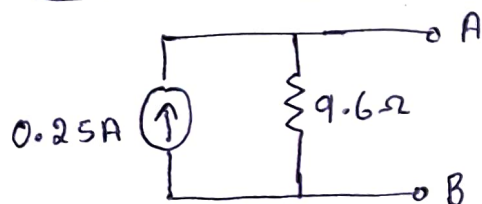
Find R_N :-

$$R_N = \frac{V_{th}}{I_N}$$

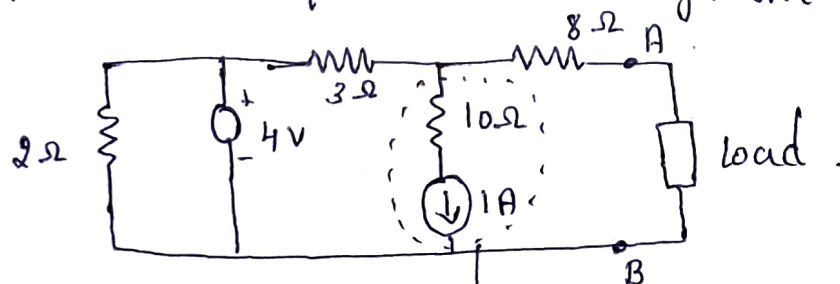
$$= \frac{2.4}{0.25}$$

$$R_N = 9.6 \Omega$$

Equivalent's Circuit :-



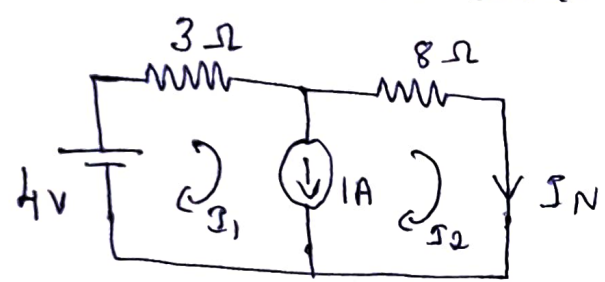
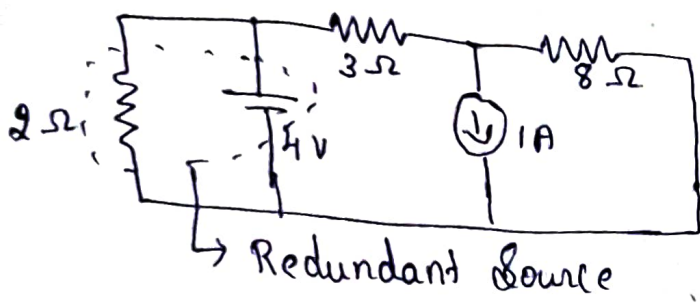
Q3. obtain Norton's equivalent circuit for the network shown.



Redundant source

Hence resistance can be neglected.

⇒ Step 1 :- Find I_N :-



Mesh (1) & (2) form Supermesh.

$$I_1 - I_2 = 1 \quad \text{--- (1)}$$

$$4 - 3I_1 - 8I_2 = 0$$

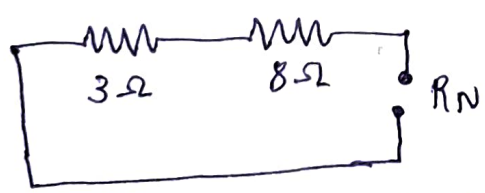
$$-3I_1 - 8I_2 = -4 \quad \text{--- (2)}$$

Solve (1) & (2) we get

$$I_1 = 1.09A \quad I_2 = 0.09A$$

$$I_2 = I_N = 0.09A$$

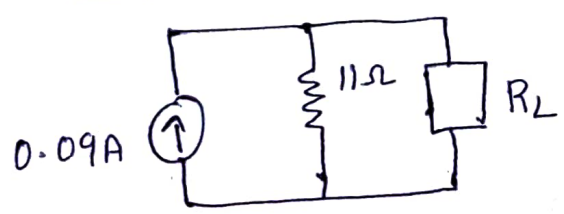
Step (2) :- To find R_N



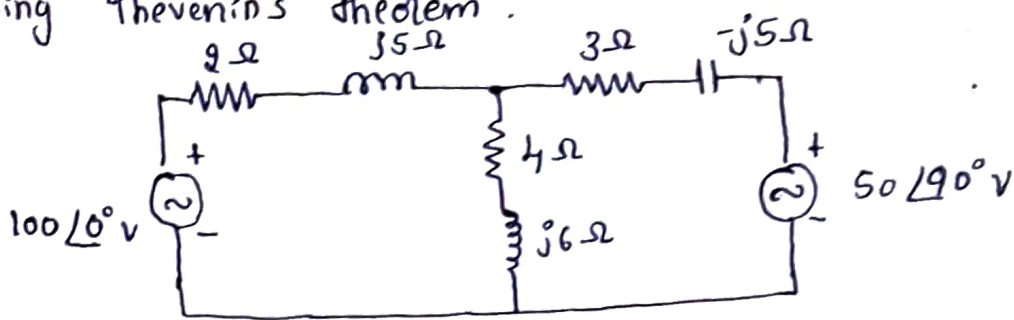
$$R_N = 3 + 8 = 11 \Omega$$

$$R_N = 11 \Omega$$

Equivalent circuit :-

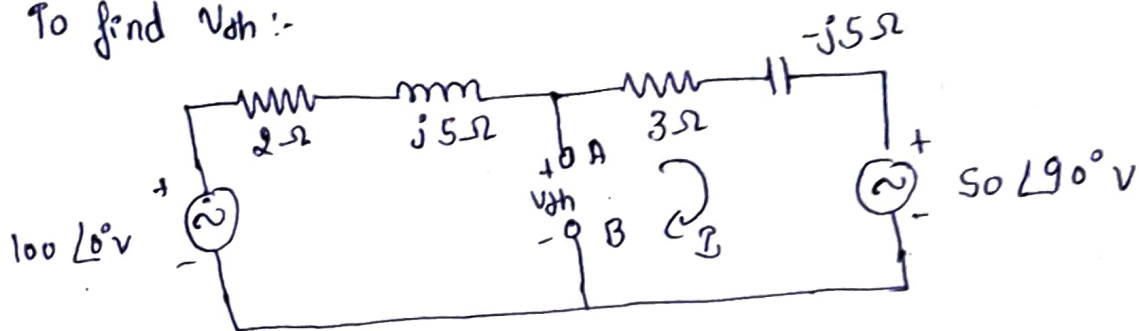


Q4 Find the current through $(4+j6)\Omega$ impedance in the network shown using Thevenin's theorem.



⇒ Step 1:-

To find V_{th} :-



Apply KVL to mesh

$$100\angle 0^\circ - 2I - j5I - 3I + j5I - 50\angle 90^\circ = 0$$

$$-5I = +50\angle 90^\circ - 100\angle 0^\circ$$

$$I = 22.36 \angle -26.57^\circ \text{ A}$$

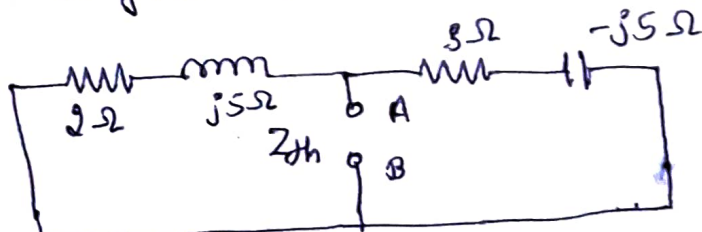
Writing V_{th} equation ,

$$V_{th} - 3I + j5I - 50\angle 90^\circ = 0$$

$$V_{th} - (3-j5)(22.36 \angle -26.57^\circ) - 50\angle 90^\circ = 0$$

$$\boxed{V_{th} = 80.61 \angle -82.88^\circ \text{ V}}$$

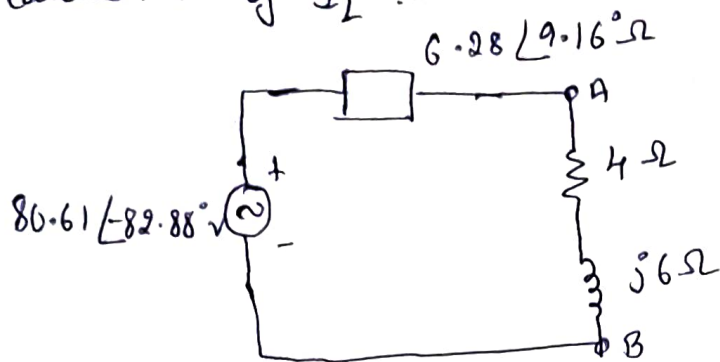
Step 2:- To find Z_{th}



$$Z_{th} = \frac{(2+j5)(3-j5)}{2+j5+3-j5}$$

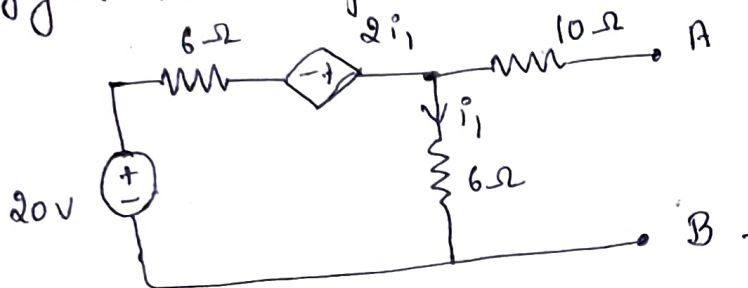
$$\boxed{Z_{th} = 6.28 \angle 9.16^\circ \Omega}$$

Q3:- Calculation of I_1 :-



$$I_1 = \frac{80.61 \angle -82.88^\circ}{6.28 \angle 9.16^\circ + 4 + j6} = 6.52 \angle -117.34^\circ \text{ A}$$

Q5. Find the thevenin & Norton equivalent for the circuit shown in figure with respect terminals A-B.



Step ①:-

To find V_{th} :-

$$20 - 6i_1 + 2i_1 - 6i_1 = 0$$

$$10i_1 = -20$$

$$i_1 = -2 \text{ A}$$

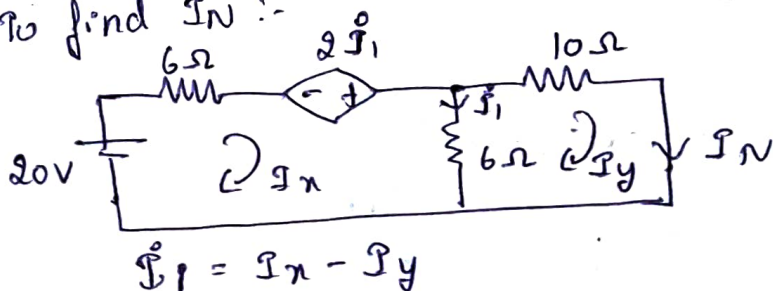
$$-10(0) - V_{th} + 6i_1 = 0$$

$$-V_{th} = -12$$

$$V_{th} = 12 \text{ V}$$

Step ②:-

To find I_N :-



Apply KVL to Mesh

Mesh (1):- $20 - 6I_x + 2I_1 - 6(I_x - I_y) = 0$
 $20 - 6I_x + 2I_x - 2I_y - 6I_x + 6I_y = 0$
 $-10I_x + 4I_y = -20 \quad \text{--- (1)}$

Mesh (2):-
 $-10I_y - 6(I_y - I_x) = 0$
 $-10I_y - 6I_y + 6I_x = 0$
 $6I_x - 16I_y = 0 \quad \text{--- (2)}$

$I_x = 2.35 \text{ A}, \quad I_y = 0.88 \text{ A}$

$I_N = I_y = 0.88 \text{ A}$

Step (3):-

To find R_N :-

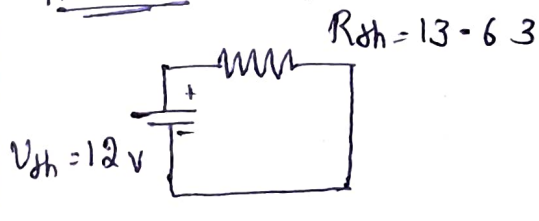
$R_N = \frac{V_{th}}{I_N} = \frac{12}{0.88} = 13.63 \Omega$

$R_{th} = R_N = 13.63 \Omega$

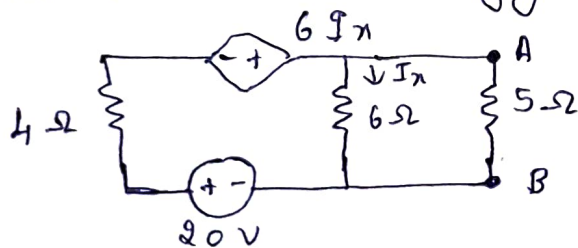
Norton's Equivalent



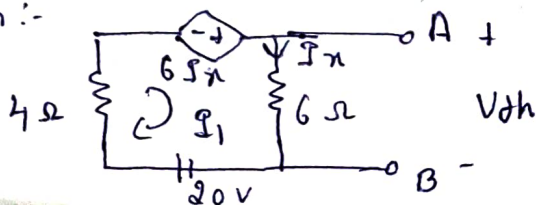
Thevenin's Equivalent



Q6. Find the current through 5Ω resistor using Norton's theorem for the circuit shown in figure.



⇒ Step (1):- To find V_{th} :-



$$20 - 4I_1 + 6I_N - 6I_1 = 0$$

$$\therefore I_1 = I_N$$

$$20 - 4I_1 + 6I_1 - 6I_1$$

$$I_1 = \frac{-20}{-4} = 5$$

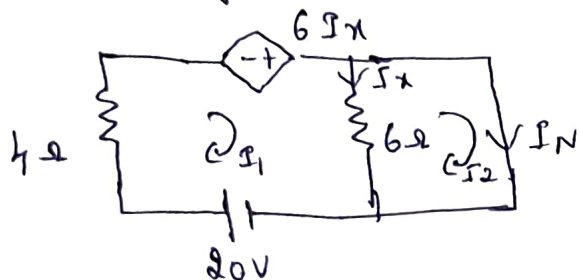
$$\therefore \boxed{I_1 = 5 \text{ A}}$$

$$-V_{th} + 6I_1 = 0$$

$$-V_{th} = -6(5)$$

$$\boxed{V_{th} = 30 \text{ V}}$$

Step 2:- To find I_N



$$I_N = I_1 - I_2$$

$$20 - 4I_1 + 6(I_1 - I_2) - 6(I_1 - I_2) = 0$$

$$20 - 4I_1 + 6I_1 - 6I_2 - 6I_1 + 6I_2 = 0$$

$$-4I_1 = -20$$

$$\boxed{I_1 = 5 \text{ A}}$$

$$-6(I_2 - I_1) = 0$$

$$-6I_2 + 6I_1 = 0$$

$$-6I_2 = -30$$

$$\boxed{I_2 = 5 \text{ A}} = I_N$$

Step 3:- To find R_N

$$R_N = \frac{V_{th}}{I_N} = \frac{30}{5} = 6 \Omega$$

Step 4:- To find I_L

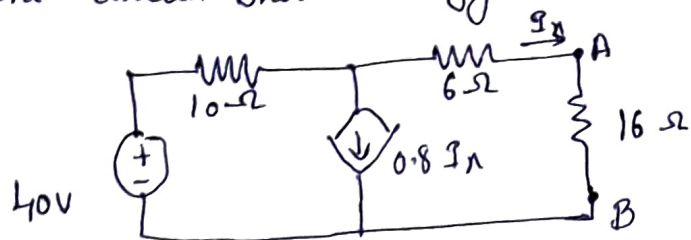


$$I_L = \frac{I_N \times R_N}{R_N + R_L}$$

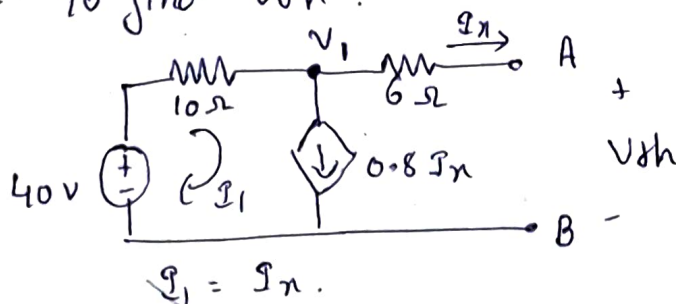
$$= \frac{5(6)}{6 + 5}$$

$$\boxed{I_L = 2.72 \text{ A}}$$

Q7. Find the current through 16Ω resistor using Norton's theorem for the circuit shown in figure.



⇒ Step (1): To find V_{th} .



$$I_1 = I_n$$

Apply KVL

$$40 - 10I_1 + 0.8I_1 = 0$$

$$-9.2I_1 = -40$$

$$I_1 = 4.34 \text{ A}$$

$$I_n = \frac{V_1}{6}$$

Applying KCL at node v_1

$$\frac{V_1 - 40}{10} + \frac{V_1}{6} + 0.8I_n = 0$$

$$\frac{V_1}{10} + \frac{V_1}{6} + 0.8 \frac{V_1}{6} = 0$$

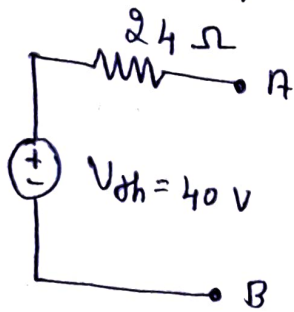
$$0.4V_1 = 4$$

$$V_1 = 10$$

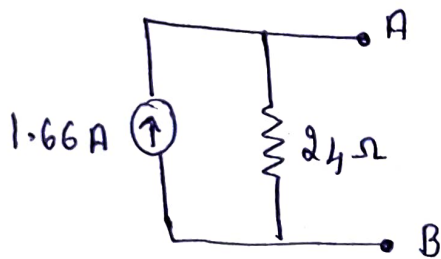
$$I_{sc} = I_n = \frac{V_1}{6} = \frac{10}{6} = 1.66 \text{ A}$$

$$Z_N = \frac{V_{oc}}{I_{sc}} = \frac{40}{1.66} = 24 \Omega$$

Thevenin's equivalent



Norton's equivalent



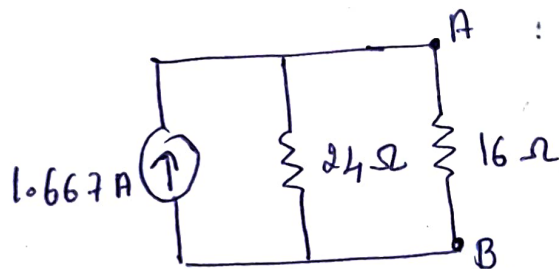
To find current through $16\ \Omega$ resistor is

$$I_{16} = 1.66 \left(\frac{24}{24 + 16} \right)$$

$$= 1.66 (0.6)$$

$$= 0.99$$

$$I_{16} \approx 1\text{ A}$$



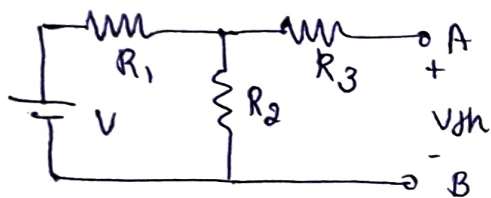
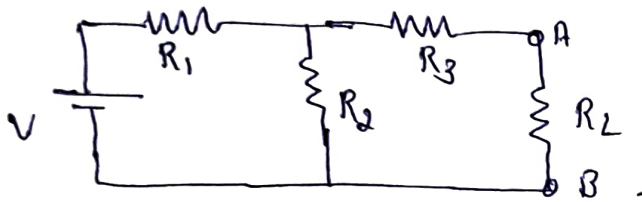
Norton's equivalent

Q8 State & prove thevenin's & Norton's theorem.

⇒ Thevenin's theorem:-

Statement :- It states that any active linear bilateral N/w connected to the load impedance may be replaced by a simple equivalent circuit containing voltage source (V_{th} & impedance R_{th} in series with it)

Proof :- Consider a simple n/w.

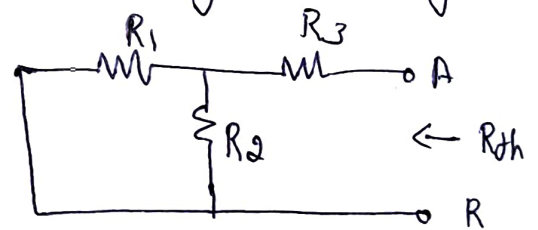


For finding load current through R_L , first remove the load resistor R_L from the n/w & calculate open circuit voltage V_{th} across point A & B

$$V_{th} = \frac{R_2}{R_1 + R_2} V$$

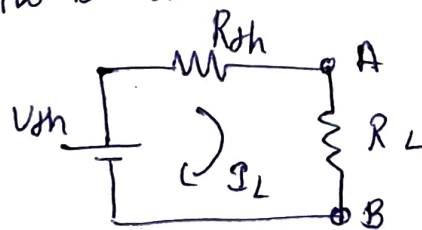
To find series resistance R_{th} , replace the voltage source by a S.C & calculate resistance b/w pt A & B

$$R_{th} = R_3 + \frac{R_1 R_2}{R_1 + R_2}$$



Thevenin's equivalent n/w is shown below.

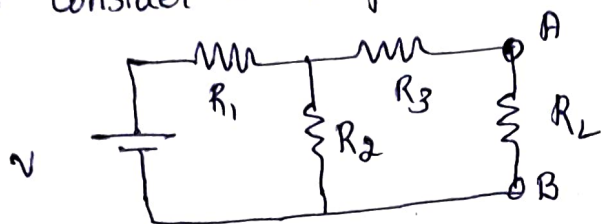
$$I_L = \frac{V_{th}}{R_{th} + R_L}$$



Norton's theorem:-

Statement:- Norton's theorem states that any active linear, bilateral N/w can be replaced by a single current source [Norton's current] in parallel with eq. Norton's resistance $[R_N]$.

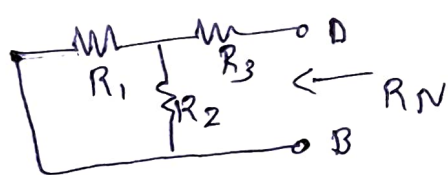
Proof:- Consider a simple N/w



To find load current through R_L , first remove the load resistor R_L from network & calculate short circuit current I_{sc} or I_N which would flow in S.C placed across terminal A & B.

To find parallel resistance R_N , replace the voltage source by a short circuit & calculate resistance b/w point A & B.

$$R_N = R_3 + \frac{R_1 R_2}{R_1 + R_2}$$



$$I_L = I_N \left(\frac{R_N}{R_N + R_L} \right)$$

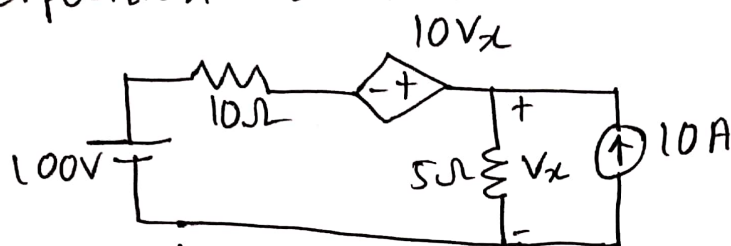
Norton's equivalent N/w.



Module-2 Unit-3

① Determine the current through 10Ω resistor using superposition theorem.

Step-I



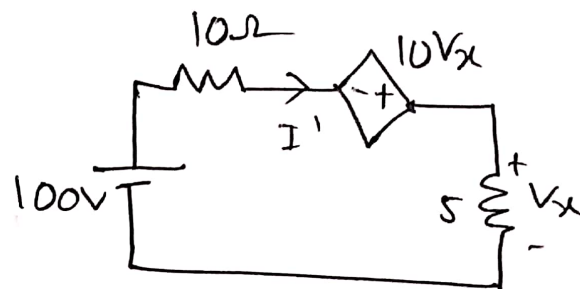
$$V_x = 5I'$$

Applying KVL to mesh

$$100 - 10I' + 10V_x - 5I' = 0$$

$$100 - 10I' + 10(5I') - 5I' = 0$$

$$I' = -2.86A$$



Step-II

when 10A is acting alone

Apply KVL to mesh 1

$$-10I_1 + 10V_x - 5(I_1 - I_2) = 0$$

$$-10I_1 + 10\{5(I_1 - I_2)\} - 5(I_1 - I_2) = 0$$

$$35I_1 - 45I_2 = 0 \quad (2)$$

Mesh 2;

$$I_2 = -10A \quad (3)$$

Solving

① & ③

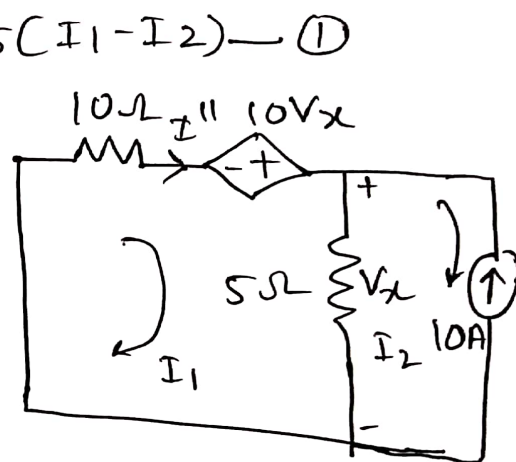
$$I_1 = -12.86A$$

$$I_2 = -10A$$

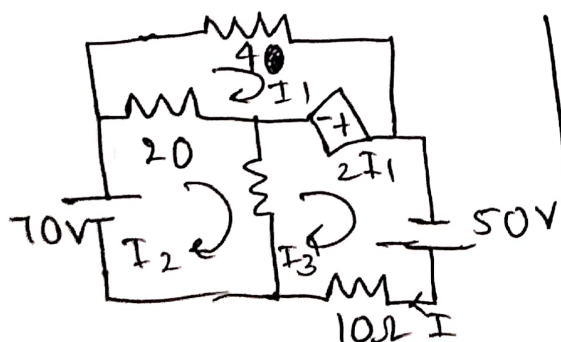
$$I'' = I_1 = -12.86A$$

By superposition

$$I = I' + I'' = -2.86 - 12.86 = 15.72A$$



② calculate the current I in n/w shown



KVL to mesh 1 $I' = I_3 - (1)$
If 70V is acting alone

$$-4I_1 - 2I_1 - 20(I_1 - I_2) = 0$$

$$26I_1 - 20I_2 = 0 \quad (2)$$

KVL to mesh 2

$$70 - 20(I_2 - I_1) - 2(I_2 - I_3) = 0$$

$$-20I_1 + 22I_2 - 2I_3 = 70 \quad (3)$$

KVL to mesh (3),

$$-2(I_3 - I_2) + 2I_1 - 10I_3 = 0$$

$$2I_1 + 2I_2 - 12I_3 = 0$$

solving

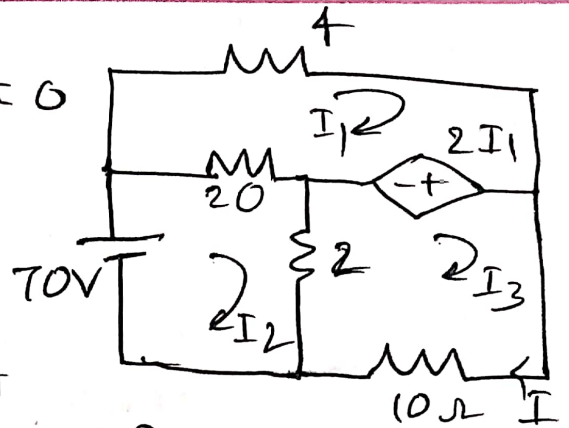
①, ②, ③ & ④

$$I_1 = 8.94 \text{ A}$$

$$I_2 = 11.62 \text{ A}$$

$$I_3 = 3.43 \text{ A}$$

$$I_1' = I_3 = 3.43 \text{ A}$$



Step: II

when 50V

source is acting alone $\rightarrow I'' = I_3$ ①

Mesh-1 $\rightarrow$

$$-4I_1 - 2I_1 - 20(I_1 - I_2) = 0$$

$$26I_1 - 20I_2 = 0 \quad \text{--- (2)}$$

Mesh: 2 $\rightarrow$

$$-20(I_2 - I_1) - 2(I_2 - I_3) = 0$$

$$-20I_1 + 22I_2 - 2I_3 = 0 \quad \text{--- (3)}$$

Mesh: 3 $\rightarrow$

$$-2(I_3 - I_2) + 2I_1 + 50 - 10I_3 = 0$$

$$2I_1 + 2I_2 - 12I_3 = -50 \quad \text{--- (4)}$$

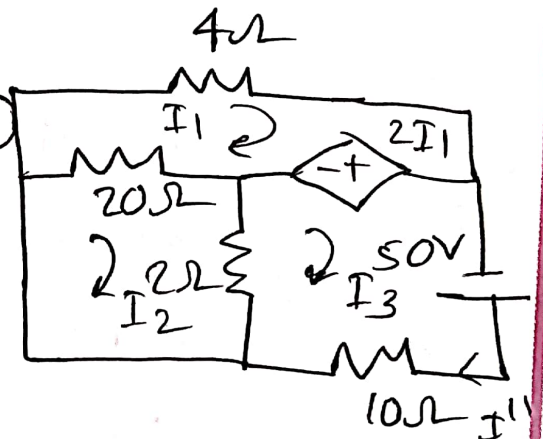
solving ①, ②, ③, ④

$$I_1 = 1.06 \text{ A}$$

$$I_2 = 1.38 \text{ A}$$

$$I_3 = 4.57 \text{ A}$$

$$I'' = I_3 = 4.57 \text{ A}$$



By superposition
Theorem

$$\rightarrow I = I' + I'' = 3.43 + 4.57 = 8 \text{ A}$$

3) state and prove Maximum power Transfer theorem for following cases:

case (i): when load impedance is variable resistance

case (ii): when load impedance is a complex impedance with variable resistance & variable reactance

→ this theorem is used to determine the value of load impedance for which the source will transfer maximum power.

consider a simple ntw as shown there are 3 possible cases for load impedance Z_L .

case (i): when load impedance is variable resistance

$$I_L = \frac{V_S}{Z_S + Z_L} = \frac{V_S}{R_S + jX_S + R_L}$$

$$|I_L| = \frac{|V_S|}{\sqrt{(R_S + R_L)^2 + X_S^2}}$$

Power delivered to load is

$$P_L = |I_L|^2 R_L = \frac{|V_S|^2 R_L}{\sqrt{(R_S + R_L)^2 + X_S^2}}$$

for power to be maximum,

$$\frac{dP_L}{dR_L} = 0$$

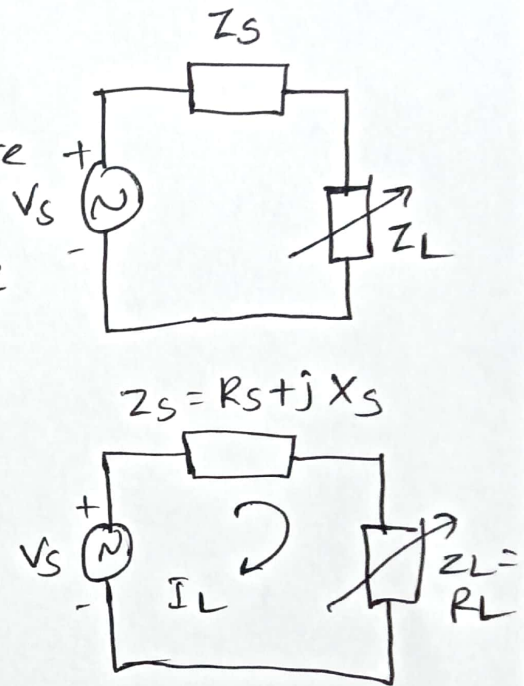
$$|V_S|^2 \left[\frac{\{ (R_S + R_L)^2 + X_S^2 \} - 2R_L(R_S + R_L)}{[(R_S + R_L)^2 + X_S^2]^2} \right] = 0$$

$$(R_S + R_L)^2 + X_S^2 - 2R_L(R_S + R_L) = 0$$

$$R_S^2 + 2R_S R_L + R_L^2 + X_S^2 - 2R_L R_S - 2R_L^2 = 0$$

$$R_S^2 + X_S^2 - R_L^2 = 0$$

$$R_L^2 = R_S^2 + X_S^2$$



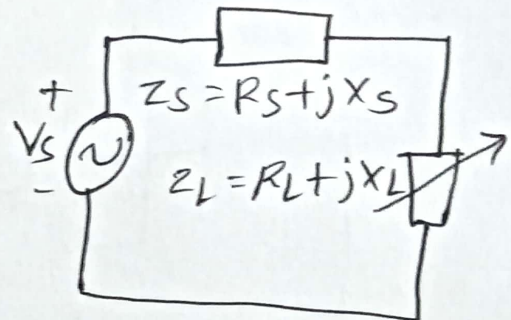
$$R_L = \sqrt{R_s^2 + X_s^2} = |Z_s|$$

Hence, load resistance R_L should be equal to magnitude of the source impedance for maximum power transfer.

case (ii) when load impedance is complex impedance with variable resistance & variable reactance

$$I_L = \frac{V_s}{Z_s + Z_L}$$

$$|I_L| = \frac{|V_s|}{\sqrt{(R_s + R_L)^2 + (X_s + X_L)^2}}$$



Power delivered to load is

$$P_L = |I_L|^2 R_L = \frac{|V_s|^2 R_L}{(R_s + R_L)^2 + (X_s + X_L)^2}$$

For max P_L , denominator of eq should be small i.e., $X_L = -X_s$

$$P_L = \frac{|V_s|^2 R_L}{(R_s + R_L)^2}$$

Differentiating above eq wrt R_L & equating to zero

$$\frac{dP_L}{dR_L} = |V_s|^2 \left[\frac{(R_s + R_L)^2 - 2R_L(R_s + R_L)}{(R_s + R_L)^4} \right] = 0$$

$$(R_s + R_L)^2 - 2R_L(R_s + R_L) = 0$$

$$R_s^2 + 2R_s R_L + R_L^2 - 2R_L R_s - 2R_L^2 = 0$$

$$R_s^2 - R_L^2 = 0$$

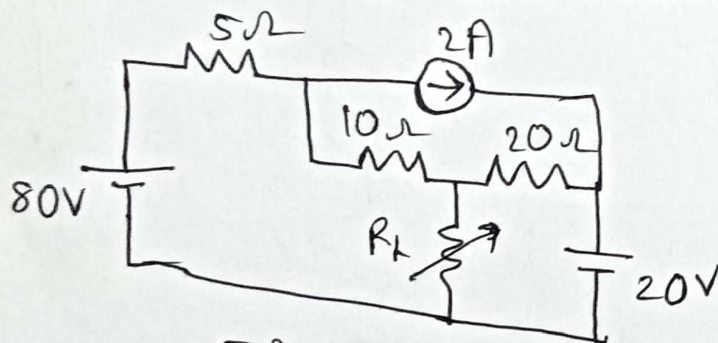
$$R_L^2 = R_S^2$$

$$R_L = R_S$$

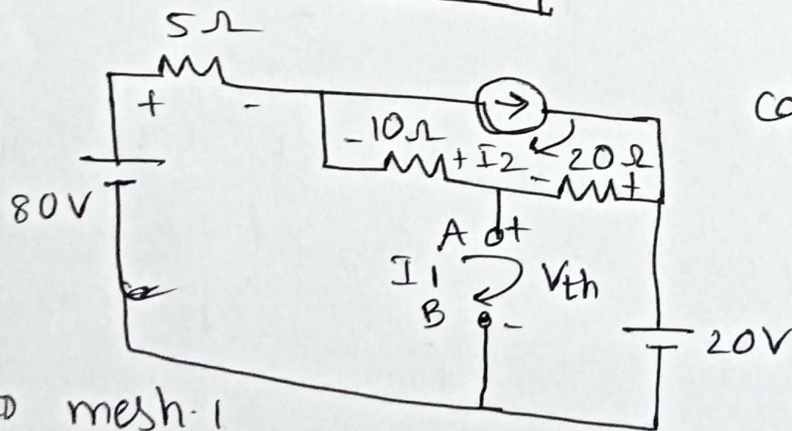
Hence, load resistance R_L should be equal to source resistance R_S & load reactance X_L should be negative value of source reactance for max power transfer.

ie load impedance should be complex conjugate of source impedance.

④ Find value of resistance R_L for maximum power transfer & calculate maximum power.



step 1:



calculating V_{th}

KVL to mesh-1

$$80 - 5I_1 - 10(I_1 - I_2) - 20(I_1 - I_2) - 20 = 0$$

$$35I_1 - 30I_2 = 60 \quad \text{--- (1)}$$

writing current eq for mesh 2

$$I_2 = 2 \quad \text{--- (2)}$$

solving (1) & (2)

$$I_1 = 3.43A$$

$$V_{th} - 20(I_1 - I_2) - 20 = 0$$

$$V_{th} = 20(3.43 - 2) + 20 = 48.6V$$

step: II
calculating R_{th}

$$R_{th} = 15 \parallel 20 = 8.57 \Omega$$

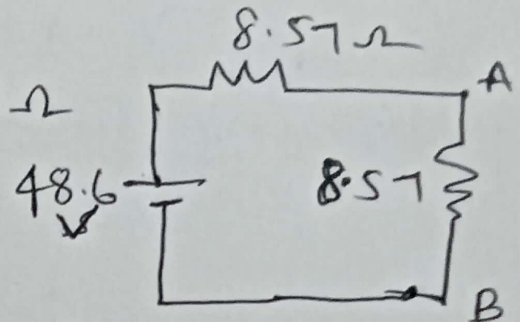
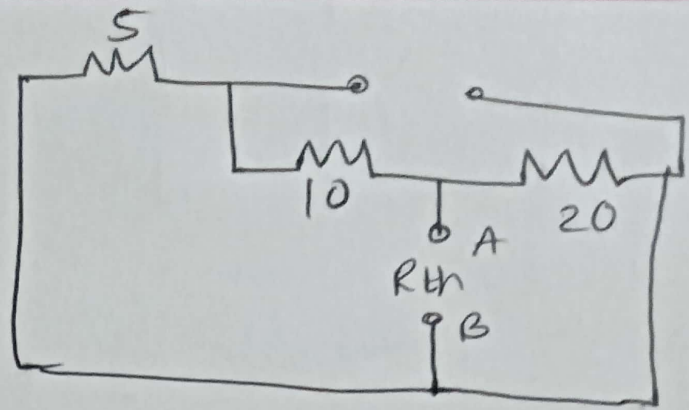
step: III
calculation
of R_L

for max
power
transfer:

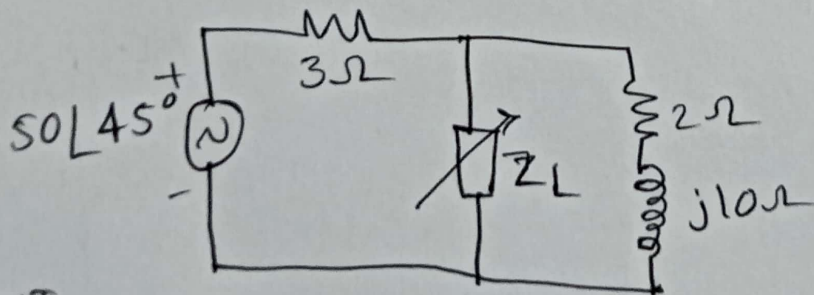
$$R_L = R_{th} = 8.57 \Omega$$

step: IV
calculation of P_{max}

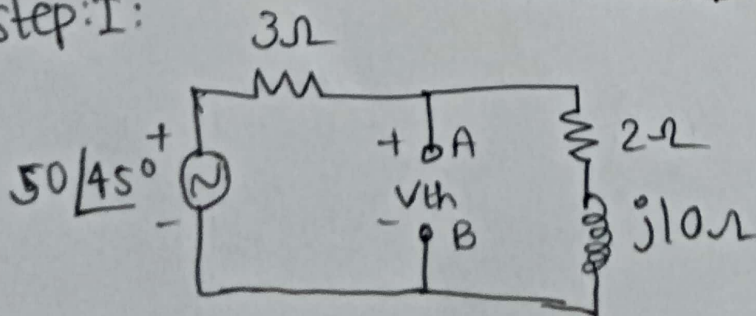
$$P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{(48.6)^2}{4(8.57)} = 68.9 \text{ W}$$



⑤ Find value of load impedance Z_L so that maximum power can be transferred to it in n/w. find max power.



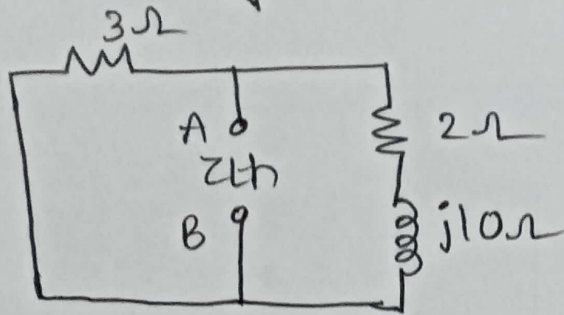
step: I:



$$I = \frac{50 \angle 45^\circ}{3 + 2 + j10} = 4.47 \angle -18.43^\circ \text{ A}$$

$$V_{th} - (2+j10)I = (2+j10)(4.47 \angle -18.43^\circ) = 45.6 \angle 60.26^\circ \text{ V}$$

Step: II: calculation of Z_{th}



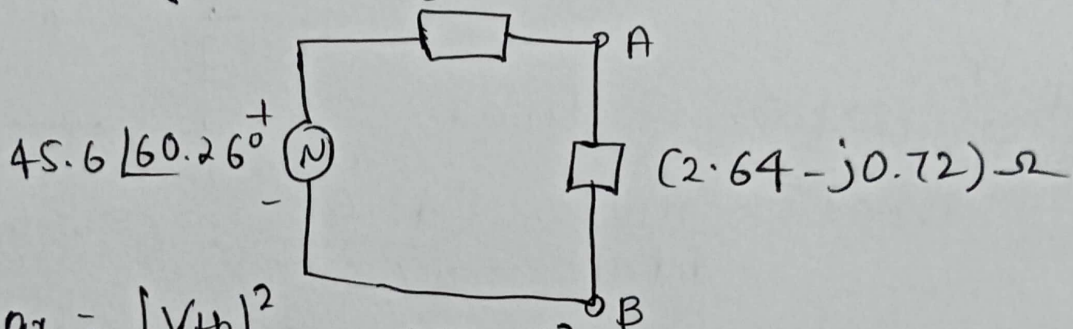
$$Z_{th} = \frac{3(2+j10)}{3+2+j10} = (2.64 + j0.72) \Omega$$

Step: III

calculation of Z_L
For maximum power transfer, load impedance should be complex conjugate

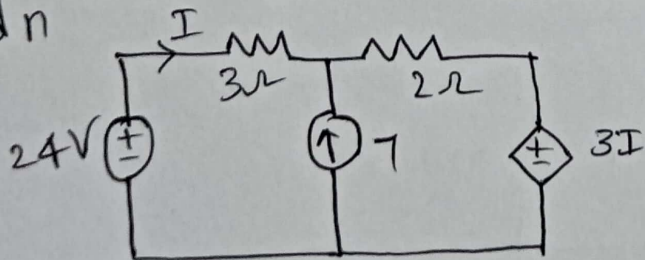
$$Z_L = (2.64 - j0.72) \Omega$$

calculation of P_{max}



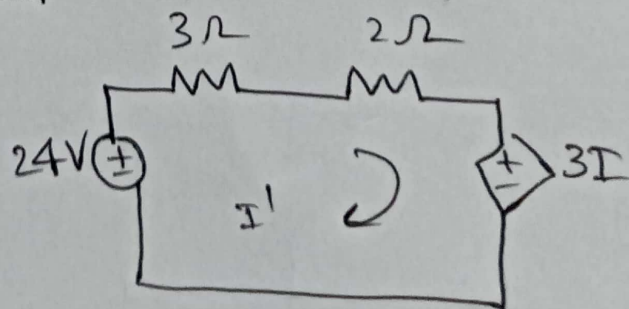
$$P_{max} = \frac{|V_{th}|^2}{4R_L} = \frac{|45.6|^2}{4 \times 2.64} = 196.9 \text{ W}$$

⑥ using super position theorem find current I in ckt shown

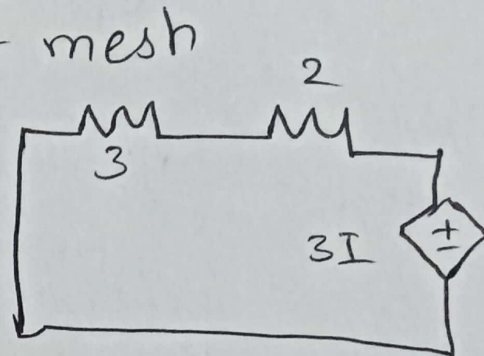
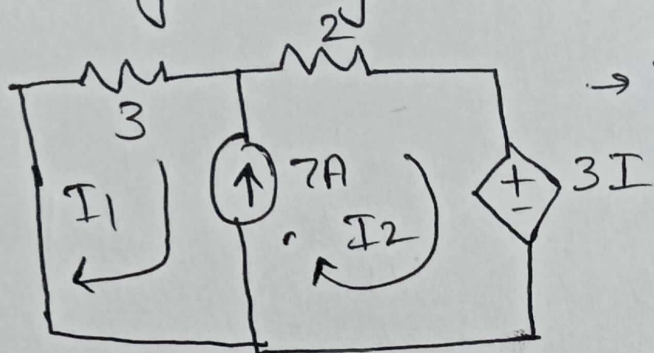


only 24V source

$$\begin{aligned} -24 + 3I' + 2I' + 3I' &= 0 \\ (3+2+3)I' &= 24 \\ 8I' &= 24 \\ I' &= 3 \text{ A} \end{aligned}$$



Activating only 7A source



$$I_2 - I_1 = 7A \quad \text{--- (1)}$$

$$-3I_1 - 2I_2 - 3I_1 = 0$$

$$-6I_1 - 2I_2 = 0$$

$$I_1 = -1.75A$$

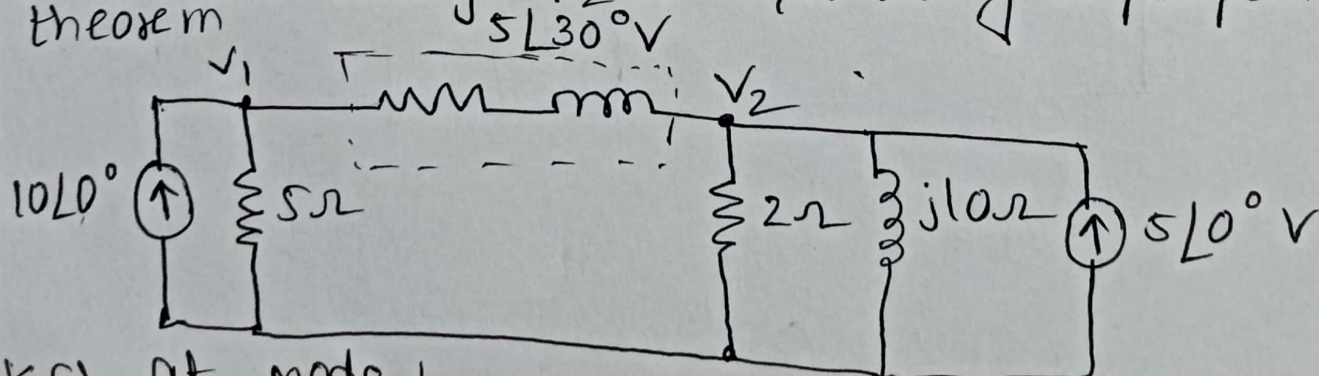
$$I_2 = 5.25A$$

$$I'' = I_1 = -1.75A$$

$$I = I' + I'' = 3 + (-1.75) = 1.25A$$

⑦

Find node voltage V_2 in n/w using superposition theorem



KCL at node 1,

$$\frac{V_1}{5} + \frac{V_1 - V_2}{5\angle 30^\circ} = 10\angle 0^\circ$$

$$\left(\frac{1}{5} + \frac{1}{5\angle 30^\circ}\right)V_1 - \frac{1}{5\angle 30^\circ}V_2 = 10\angle 0^\circ =$$

KCL at node 2 $(0.37 - j0.1)V_1 - (0.17 - j0.1)V_2 = 10\angle 0^\circ \quad (1)$

$$\frac{V_2 - V_1}{5\angle 30^\circ} + \frac{V_2}{2} + \frac{V_2}{j10} = 0$$

$$-\frac{1}{5\angle 30^\circ} V_1 + \left[\frac{1}{5\angle 30^\circ} + \frac{1}{2} + \frac{1}{j10} \right] V_2 = 0$$

$$-(0.17 - j0.1)V_1 + (0.67 - j0.2)V_2 = 0 \quad (2)$$

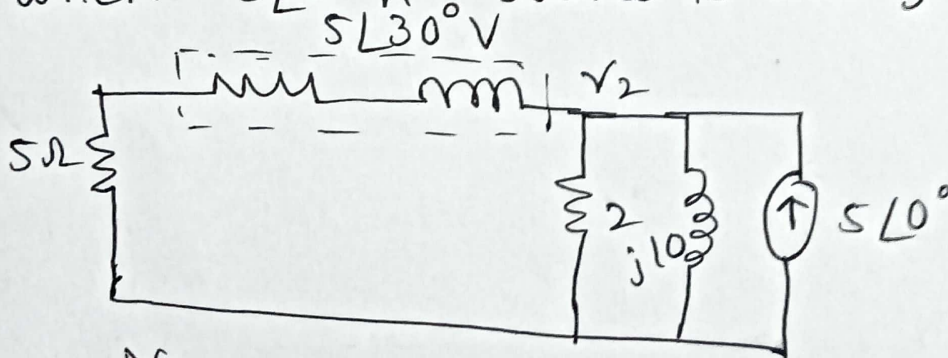
Eq (1) & (2) in matrix form

$$\begin{bmatrix} 0.37 - j0.1 & -(0.17 - j0.1) \\ -(0.17 - j0.1) & 0.67 - j0.2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 10\angle 0^\circ \\ 0 \end{bmatrix}$$

By Cramer's rule;

$$V_2 = \frac{\begin{vmatrix} 0.37 - j0.1 & 10\angle 0^\circ \\ -(0.17 - j0.1) & 0 \end{vmatrix}}{\begin{vmatrix} 0.37 - j0.1 & -(0.17 - j0.1) \\ -(0.17 - j0.1) & 0.67 - j0.2 \end{vmatrix}} = 8.57\angle -3.36^\circ \text{ V}$$

step: 2 when $5\angle 0^\circ \text{ A}$ source is acting alone



$$\frac{V_2}{5\angle 30^\circ + 5} + \frac{V_2}{2} + \frac{V_2}{j10} = 5\angle 0^\circ$$

$$(0.61\angle -11.93^\circ)V_2 = 5\angle 0^\circ$$

$$V_2 = 8.2\angle 11.93^\circ \text{ V}$$

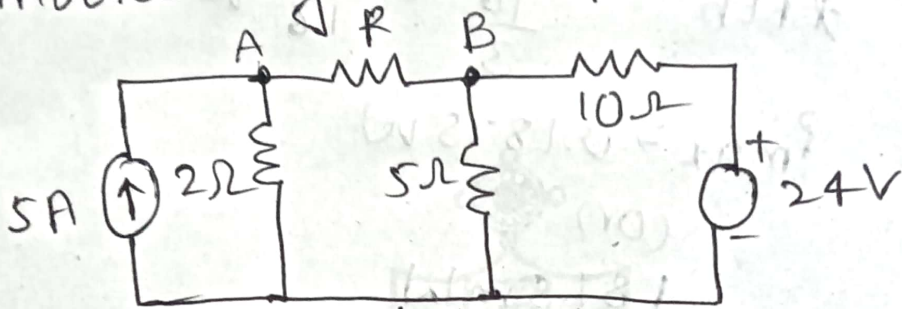
By
superposition
theorem

$$V_2 = V_2' + V_2'' = 8.57\angle -3.36^\circ + 8.2\angle 11.93^\circ = 16.62\angle 4.12^\circ \text{ V}$$

START WRITING FROM HERE

Write 25 lines in each page

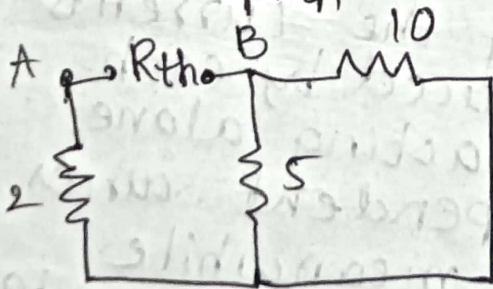
state maximum, power transfer theorem for ckt shown, what should be value of R such that maximum power transfer can take place from rest of n/w. obtain amount of this power.



In a linear, bilateral DC n/w, maximum power is delivered from source n/w to a variable resistance (R_L) when load resistance is equal to internal resistance.

$$R_L = R_{th}$$

$$P_{max} = \frac{V_{th}^2}{4R_{th}}$$



$$R_B = 5 \parallel 10 = \frac{5 \times 10}{5 + 10} = \frac{50}{15} = 3.33$$

$$R_{th} = R_A + R_B = 2 + 10/3 = 5.33 \Omega$$

To find V_{th}

$$At \quad V_A = 5A \times 2\Omega = 10V$$

$$V_B = 24V \times \frac{5}{5+10} = 24 \times \frac{5}{15} = 24 \times \frac{1}{3} = 8V$$

24V source forms simple voltage divider ckt across series combination of 10Ω & 5Ω resistors.

$$V_{th} = V_A - V_B$$

$$= 10 - 8 = 2V$$

$$P_{max} = \frac{V_{th}^2}{4R_{th}}$$

$$= \frac{2^2}{4 \times \frac{16}{3}} = \frac{1}{\frac{16}{3}} = \frac{3}{16} W$$

$$P_{max} = 0.1875 W$$

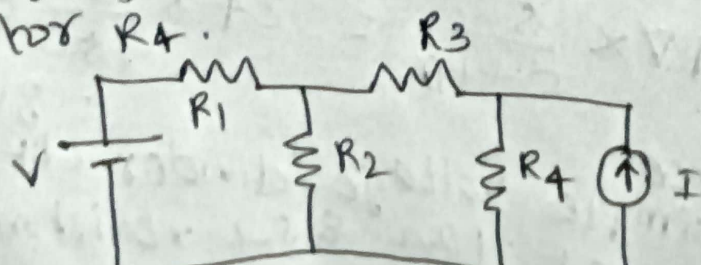
(or)

$$187.5 mW$$

⑨ state and prove superposition theorem.
It states that

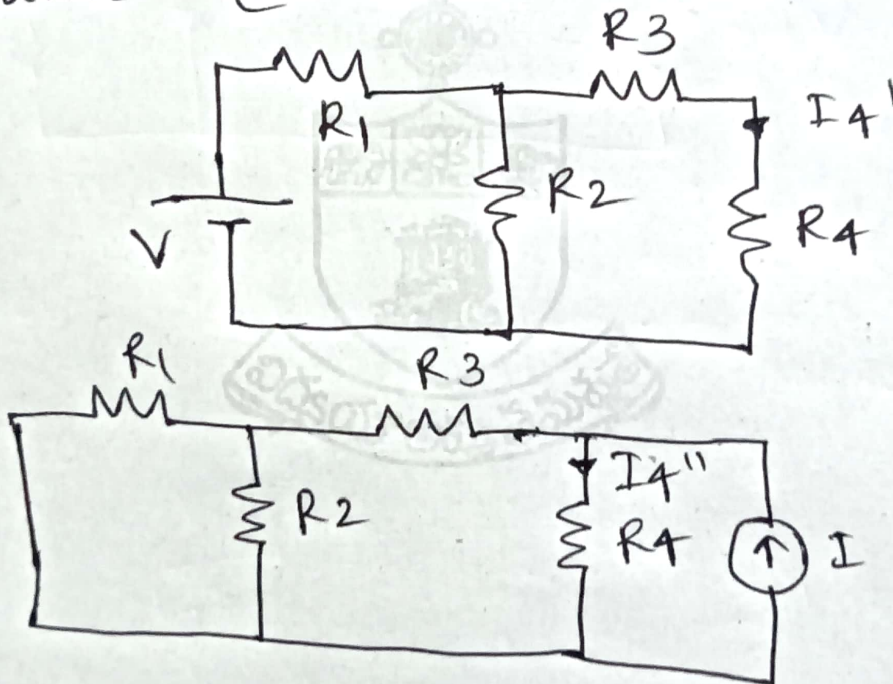
In a linear network containing more than one independent source & dependent source, the resultant current in any element is the algebraic sum of the currents that would be produced by each independent source acting alone, all the other independent sources being represented meanwhile by their respective internal resistances.

consider n/w show. suppose we have to find current I_4 through resistor R_4 .



The current flowing through R_4 due to constant voltage source V is found to be say I_4' , representing constant current source with ∞ infinite resistance, i.e. open circuit.

The current flowing through resistor R_4 due to constant current source I is found to be say I_4'' , representing the constant voltage source with zero resistance (or short circuit).



The resultant current I_4 through R_4 is found by superposition theorem

$$I_4 = I_4' + I_4''$$